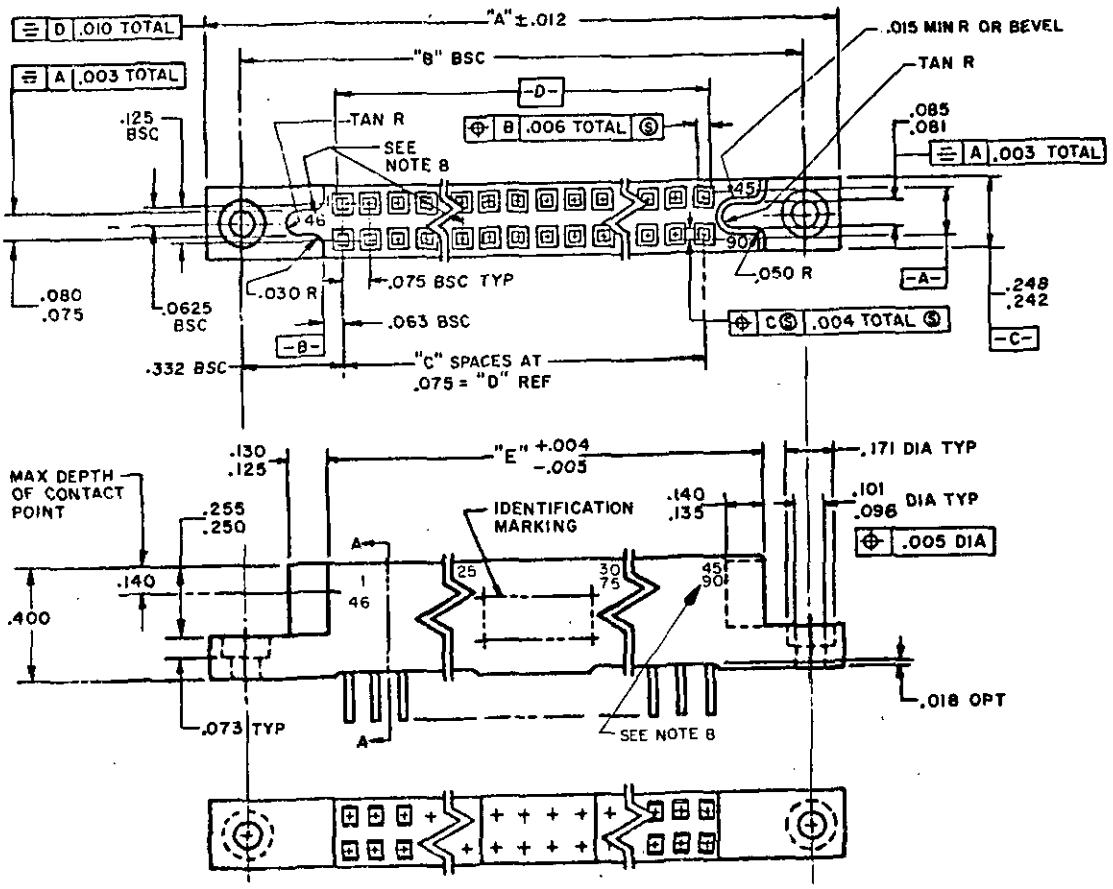
MIL-C-55302/240
24 March 1980
SUPERSEDING
MIL-C-55302/24C
20 March 1972

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, DECADE INCREMENT 10 THRU 110 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.075 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-55302.



Ⓓ STYLE I (INTERNAL KEYING)

FIGURE 1. Connectors, receptacle (.075 spacing).

Ⓓ denotes changes

FSC 5935

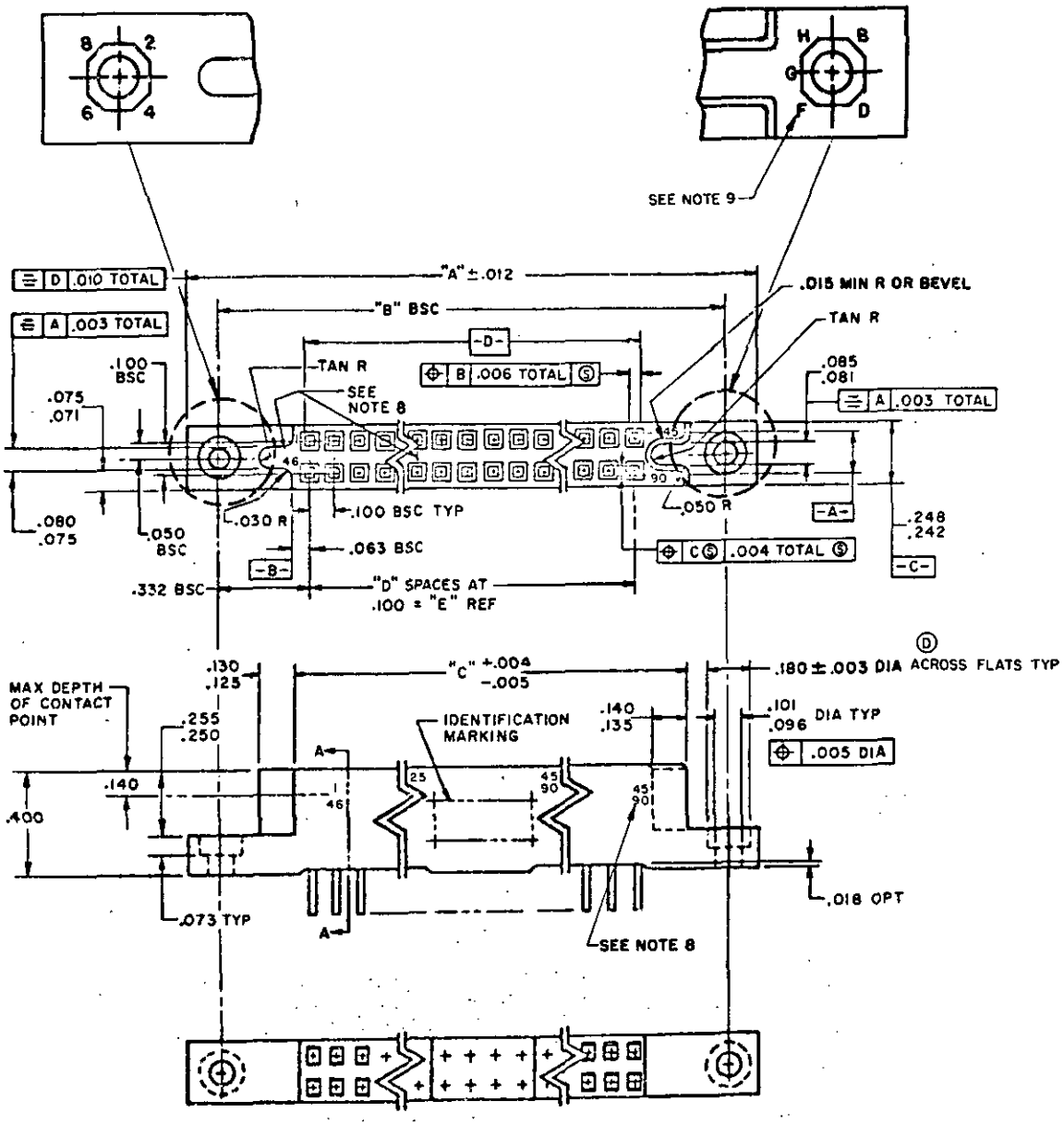
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1/ Dash number	No. of contacts	Type	A	B	C	D	E	Contact identification numbers
01		I						
02	10	II	1.255	.965	4	.300	.565	1 - 5
03		III	(31.88)	(24.51)		(7.62)	(14.35)	6 - 10
04		IV						
05		I						
06	20	II	1.630	1.340	9	.675	.940	1 - 10
07		III	(41.40)	(34.04)		(17.15)	(23.88)	11 - 20
08		IV						
09		I						
10	30	II	2.005	1.715	14	1.050	1.315	1 - 15
11		III	(50.93)	(43.56)		(26.67)	(33.40)	16 - 30
12		IV						
13		I						
14	40	II	2.380	2.090	19	1.425	1.690	1 - 20
15		III	(60.45)	(53.09)		(36.20)	(42.93)	21 - 40
16		IV						
17		I						
18	50	II	2.755	2.465	24	1.800	2.065	1 - 25
19		III	(69.98)	(62.61)		(45.72)	(52.45)	26 - 50
20		IV						
21		I						
22	60	II	3.130	2.840	29	2.175	2.440	1 - 30
23		III	(79.50)	(72.14)		(55.25)	(61.98)	31 - 60
24		IV						
25		I						
26	70	II	3.505	3.215	34	2.550	2.815	1 - 35
27		III	(89.03)	(81.66)		(64.77)	(71.50)	36 - 70
28		IV						
29		I						
30	80	II	3.880	3.590	39	2.925	3.190	1 - 40
31		III	(98.55)	(91.19)		(74.30)	(81.03)	41 - 80
32		IV						
33		I						
34	90	II	4.255	3.965	44	3.300	3.565	1 - 45
35		III	(108.08)	(100.71)		(83.82)	(90.55)	46 - 90
36		IV						
37		I						
38	100	II	4.630	4.340	49	3.675	3.940	1 - 50
39		III	(117.60)	(110.24)		(93.35)	(100.08)	51 - 100
40		IV						
41		I						
42	110	II	5.005	4.715	54	4.050	4.315	1 - 55
43		III	(127.13)	(119.76)		(102.87)	(109.60)	56 - 110
44		IV						

Ⓓ 1/ Dash numbers 01 thru 44 are for style 1 (internal keying) connectors.

FIGURE 1. Connectors, receptacle, (.075 spacing) - Continued.

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STYLE 2 (EXTERNAL KEYING)

D FIGURE 1. Connectors, receptacle (.075 spacing) - Continued.

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Ⓓ

1/ Dash number	No. of contacts	Type	A	B	C	D	E	Contact identification numbers
45	10	I	1.255 (31.88)	.965 (24.51)	4	.300 (7.62)	.565 (14.35)	1 - 5
46		II						6 - 10
47		III						
48		IV						
49	20	I	1.630 (41.40)	1.340 (34.04)	9	.675 (17.15)	.940 (23.88)	1 - 10
50		II						11 - 20
51		III						
52		IV						
53	30	I	2.005 (50.93)	1.715 (43.56)	14	1.050 (26.67)	1.315 (33.40)	1 - 15
54		II						16 - 30
55		III						
56		IV						
57	40	I	2.380 (60.45)	2.090 (53.09)	19	1.425 (36.20)	1.690 (42.93)	1 - 20
58		II						21 - 40
59		III						
60		IV						
61	50	I	2.755 (69.98)	2.465 (62.61)	24	1.800 (45.72)	2.065 (52.45)	1 - 25
62		II						26 - 50
63		III						
64		IV						
65	60	I	3.130 (79.50)	2.840 (72.14)	29	2.175 (55.25)	2.440 (61.98)	1 - 30
66		II						31 - 60
67		III						
68		IV						
69	70	I	3.505 (89.03)	3.215 (81.66)	34	2.550 (64.77)	2.815 (71.50)	1 - 35
70		II						36 - 70
71		III						
72		IV						
73	80	I	3.880 (98.55)	3.590 (91.19)	39	2.925 (74.30)	3.190 (81.03)	1 - 40
74		II						41 - 80
75		III						
76		IV						
77	90	I	4.255 (108.08)	3.965 (100.71)	44	3.300 (83.82)	3.565 (90.55)	1 - 45
78		II						46 - 90
79		III						
80		IV						
81	100	I	4.630 (117.60)	4.340 (110.24)	49	3.675 (93.35)	3.940 (100.08)	1 - 50
82		II						51 - 100
83		III						
84		IV						
85	110	I	5.005 (127.13)	4.715 (119.76)	54	4.050 (102.87)	4.315 (109.60)	1 - 55
86		II						56 - 110
87		III						
88		IV						

1/ Dash numbers 45 thru 88 are for style 2 (external keying) connectors.

Ⓓ FIGURE 1. Connectors, receptacle, (.075 spacing) - Continued.

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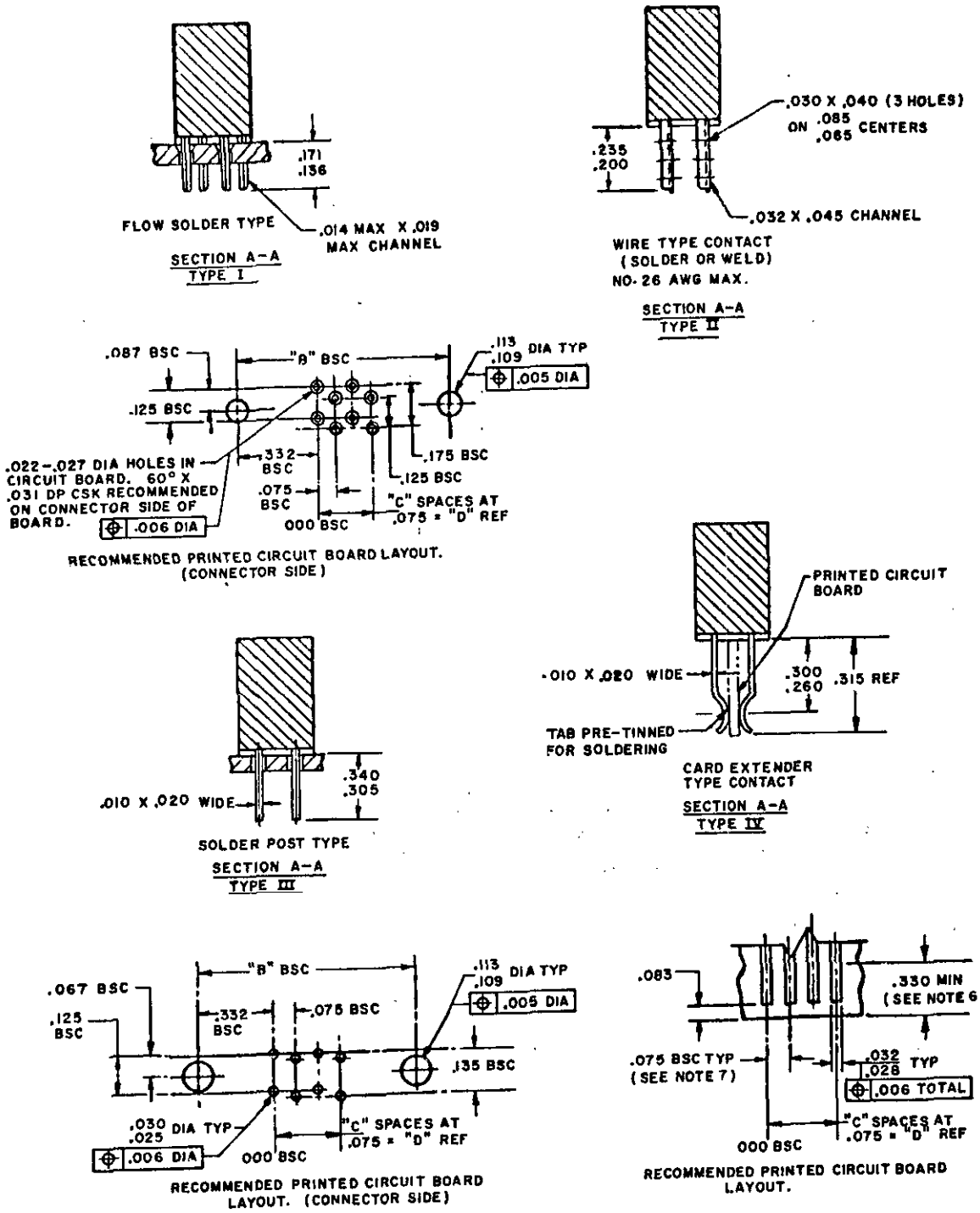


FIGURE 1. Connectors, receptacle (.075 spacing). - Continued

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INCHES	MM	INCHES	MM
.003	.08	.083	2.11
.004	.10	.085	2.16
.005	.13	.087	2.21
.006	.15	.096	2.44
.010	.25	.101	2.57
.012	.30	.109	2.77
.014	.36	.113	2.87
.015	.38	.125	3.18
.018	.46	.130	3.30
.019	.48	.135	3.43
.020	.51	.136	3.45
.022	.56	.140	3.56
.025	.64	.171	4.34
.027	.69	.175	4.44
.028	.71	.180	4.57 (D)
.030	.76	.200	5.08
.031	.79	.235	5.97
.032	.81	.242	6.15
.040	1.02	.248	6.30
.045	1.14	.250	6.35
.050	1.27	.255	6.48
.0625	1.588	.260	6.60
.063	1.60	.300	7.62
.065	1.65	.305	7.75
.067	1.70	.315	8.00
.073	1.85	.330	8.38
.075	1.90	.332	8.43
.080	2.03	.340	8.64
.081	2.06	.400	10.16

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/23.
5. Use printed circuit boards with the Type III receptacle.
6. Minimum pad length on Printed Circuit board.
7. Termination layout on .025 (.64 mm) modular grid.
8. Numbers indicating, every five cavities stamped on side (except 10 position). Numbers indicating end cavities and lines indicating every five cavities stamped on mating face.
(D) 9. Location indicators embossed on surface.

FIGURE 1. Connectors, receptacle, (.075 spacing) - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Material:
Plating: The contact plating shall be gold per MIL-G-45204, type II, grade C, class I, over nickel per QQ-N-290, class 2, 30 to 150 microinches.

Contact identification: See figure 1.

Contact rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per connector at room ambient.

Keying: See MIL-C-55302/31.

Oversized pin exclusion (sockets only): .0465 diameter pin.

Contact separation force: The connector shall meet the requirements of MIL-C-55302/29.

Wire size: No. 26 AWG maximum with type II, solder eyelet contact for wire termination.

Resistance to test probe damage: The connector shall meet the requirements of MIL-C-55302/30.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .25 and the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: 3 pounds minimum.

Dielectric withstanding voltage:
Sea level: 900 volts rms.
High altitude: 200 volts rms.

Replacement contacts: See MIL-C-55302/32.

Contact removal tool: See MIL-C-55302/28.

Part number: M55302/24- (dash number from figure 1).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - CR, AR, MI
Navy - SH, AS, OS
Air Force - 11, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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